

[BANK NAME] Credit and Underwriting Policy Manual

Section 7: Automated Lending Decisioning Standards A Customisable Template for AI-Driven Loan Origination Systems

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This is a **policy template**. Every field marked **[TEMPLATE VALUE]** must be replaced with this institution's board-approved figures before this document is published internally or ingested into the AI decisioning RAG knowledge base. Changes to any threshold require sign-off from the Chief Credit Officer and notification to the Board Risk Committee.

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7.1 Policy Scope and Governance

This section governs the operation of the bank's Automated Decisioning System (ADS). The ADS processes retail and MSME credit applications using a multi-node AI pipeline that analyses bureau data, income information, and behavioural signals to produce an APPROVE, COUNTER_OFFER, or DECLINE output.

7.1.1 Governance Structure

Role	Responsibility
Chief Credit Officer	Owns this policy; approves all threshold changes; signs model governance reviews
Board Risk Committee	Receives quarterly model performance reports; approves major policy revisions
Technology Risk Team	Monitors FALLBACK rate, model drift, and RAG retrieval quality
Compliance Officer	Ensures RBI obligations are met; manages audit log access
Credit Operations Team	Reviews FALLBACK-path decisions within 24 hours; processes human overrides

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7.2 Data Handling and PII Masking Policy

ADS node: `pi_deletion_node` (mandatory pipeline entry point).

7.2.1 Policy Statement

All credit bureau payloads must be sanitised of Personally Identifiable Information before entering any AI or ML processing step. The `pi_deletion_node` is the mandatory first step. No downstream node may access the raw bureau payload. A deep copy must be created before masking — in-place mutation is prohibited.

Data Category	Fields to Remove	Method
Consumer Names	firstName, middleName, surname	Delete key from payload
Address	streetPrefix, streetName, streetSuffix, unitType, unitId, city, state, zipCode	Delete key from payload
Employment	Employer name in tradeline	Delete key from payload
Account Numbers	accountNumber on all tradelines	Delete key from payload
Court Records	courtName, referenceNumber in public records	Delete key from payload

[TEMPLATE VALUE] PII Masking Confirmation

State variable name that confirms masking is complete. Example: `is_pii_masked = True` must be set in pipeline state before any downstream node reads bureau data.

7.3 Credit Score Band and Base Limit Policy

ADS node: `credit_score_node`.

7.3.1 Policy Statement

Every applicant must be classified into a credit score band. The band determines the maximum base lending limit before any adjustment factors are applied. All thresholds below are template values and must be replaced with board-approved figures before ingesting into the RAG store.

Band	Score Range [TEMPLATE VALUE]	Max Base Limit [TEMPLATE VALUE]	Risk Flag	Aggregation Weight [TEMPLATE VALUE]
PRIME	750 or above	INR 75,00,000	LOW	0.25
NEAR_PRIME	700 to 749	INR 50,00,000	MODERATE	0.25
FAIR	650 to 699	INR 35,00,000	MODERATE	0.25
SUBPRIME	Below 650	INR 20,00,000	HIGH	0.25

[TEMPLATE VALUE] Score Band Thresholds and Base Limits

PRIME: `score_band = PRIME` when `score >= 750` — `base_limit_band = INR 75,00,000` — `score_risk_flag = LOW` — `score_weight = 0.25`. NEAR_PRIME: `score_band = NEAR_PRIME` when `score 700 to 749` — `base_limit_band = INR 50,00,000` — `score_risk_flag = MODERATE` — `score_weight = 0.25`. FAIR: `score_band = FAIR` when `score 650 to 699` — `base_limit_band = INR 35,00,000` — `score_risk_flag = MODERATE` — `score_weight = 0.25`. SUBPRIME: `score_band = SUBPRIME` when `score below 650` — `base_limit_band = INR 20,00,000` — `score_risk_flag = HIGH` — `score_weight = 0.25`. Score source: `riskModel[0].score` from `pi_masked_experian_data`. When `llm_response_type = RAG`, thresholds retrieved from knowledge base take precedence over these defaults.

7.4 Public Record Severity and Hard-Decline Policy

ADS node: `public_record_node`.

7.4.1 Policy Statement

Public record events are evaluated for severity. Each severity level carries an adjustment factor that reduces the base lending limit. SEVERE events trigger a Hard Decline flag that supersedes all other signals.

Severity	Trigger Conditions [TEMPLATE VALUE]	Adj Factor [TEMPLATE VALUE]	hard_decline_flag
NONE	No public records on file	1.00	False
LOW	Non-bankruptcy adverse records (liens, minor judgments)	0.90	False
MODERATE	Bankruptcy discharged 5 or more years ago	0.75	False
SEVERE	Bankruptcy under 5 years; multiple judgments; active proceedings	0.50	True

[TEMPLATE VALUE] Public Record Adjustment Factors and Hard-Decline Triggers

NONE: no records — `public_record_adjustment_factor` = 1.00, `hard_decline_flag` = False. LOW: non-bankruptcy adverse records — `public_record_adjustment_factor` = 0.90, `hard_decline_flag` = False. MODERATE: bankruptcy discharged $\geq$ 5 years — `public_record_adjustment_factor` = 0.75, `hard_decline_flag` = False. SEVERE: bankruptcy within 5 years OR multiple judgments OR active insolvency — `public_record_adjustment_factor` = 0.50, `hard_decline_flag` = True. Additional hard-decline triggers: bankruptcy within 24 months sets `hard_decline_flag` = True. Wilful defaulter listing sets `hard_decline_flag` = True. Active IBC Section 14 moratorium sets `hard_decline_flag` = True.

IMPORTANT: `hard_decline_flag` = True forces `aggregated_risk_score` to 0.0. This cannot be overridden by the decision node, counter-offer node, or any operator.

7.5 Credit Utilisation Adjustment Policy

ADS node: utilization_node.

7.5.1 Policy Statement

Revolving credit utilisation is computed using only revolving accounts (revolvingOrInstallment = R). The ratio determines a utilisation_adjustment_factor applied multiplicatively to the base lending limit.

Utilisation Range [TEMPLATE VALUE]	Risk Level	Adj Factor [TEMPLATE VALUE]	Interpretation
0% to 30%	EXCELLENT	0.975	Low revolving burden
31% to 50%	GOOD	0.875	Moderate; manageable
51% to 75%	MODERATE	0.725	Elevated; nearing stress
Above 75%	CRITICAL	0.575	Near-maxed; strong stress indicator

[TEMPLATE VALUE] Utilisation Thresholds and Adjustment Factors

EXCELLENT: utilisation_ratio 0.00 to 0.30 — utilisation_adjustment_factor = 0.975. GOOD: utilisation_ratio 0.31 to 0.50 — utilisation_adjustment_factor = 0.875. MODERATE: utilisation_ratio 0.51 to 0.75 — utilisation_adjustment_factor = 0.725. CRITICAL: utilisation_ratio above 0.75 — utilisation_adjustment_factor = 0.575. Formula: utilisation_ratio = total_revolving_balance divided by total_revolving_credit_limit. Only accounts with revolvingOrInstallment = R are included in the calculation.

7.6 Debt Exposure and Leverage Policy

ADS node: `debt_exposure_node`.

7.6.1 Policy Statement

Total outstanding debt across all open tradelines (`openOrClosed = O`) is assessed to ensure the applicant's leverage does not exceed the bank's risk appetite. Monthly obligations from this node feed the DTI calculation.

Exposure Risk	Total Debt Multiple [TEMPLATE VALUE]	Monthly Obligation Ceiling [TEMPLATE VALUE]	Lending Impact
LOW	Below 2 times annual income	Below 15% of monthly income	Full capacity available
MODERATE	2 to 4 times annual income	15% to 30% of monthly income	Moderate capacity reduction
HIGH	4 to 6 times annual income	30% to 45% of monthly income	Significant reduction; monitoring
EXTREME	Above 6 times annual income	Above 45% of monthly income	Counter-offer required

[TEMPLATE VALUE] Debt Exposure Risk Thresholds

LOW: `total_existing_debt` below 2 times gross annual income — `exposure_risk = LOW`. MODERATE: total debt 2 to 4 times annual income — `exposure_risk = MODERATE`. HIGH: total debt 4 to 6 times annual income — `exposure_risk = HIGH`. EXTREME: total debt above 6 times annual income — `exposure_risk = EXTREME`. `monthly_obligation_estimate` = sum of `monthlyPaymentAmount` from all open tradelines.

7.7 Inquiry Velocity and Fraud Control Policy

ADS node: inquiry_node.

7.7.1 Policy Statement

Hard credit inquiry count over a 12-month rolling window is monitored. High velocity signals distress or fraud. An inquiry_penalty_factor is applied multiplicatively to the lending limit.

Inquiry Count (12 months) [TEMPLATE VALUE]	Velocity Risk	Penalty Factor [TEMPLATE VALUE]	Required Action
0 to 1	LOW	1.00	Normal processing
2 to 3	MODERATE	0.90	Note sources in credit file
4 to 6	HIGH	0.70	Enhanced due diligence; senior review
7+	CRITICAL	0.50	Fraud screening mandatory

[TEMPLATE VALUE] Inquiry Velocity Thresholds

LOW: 0 to 1 hard inquiries in 12 months — inquiry_penalty_factor = 1.00. MODERATE: 2 to 3 inquiries — inquiry_penalty_factor = 0.90. HIGH: 4 to 6 inquiries — inquiry_penalty_factor = 0.70. CRITICAL: 7 or more inquiries — inquiry_penalty_factor = 0.50. Soft inquiries excluded. Penalty factor applied multiplicatively after other adjustments.

7.8 Payment Behaviour Scoring Policy

ADS node: `payment_behavior_node`.

7.8.1 Policy Statement

Applicants are scored on a 0 to 100 behaviour scale derived from delinquency counts and chargeoff history across all tradelines. The `behaviour_score` feeds directly into risk aggregation at the weight defined in Section 7.11.

Risk Band	30+ DPD Count [TEMPLATE VALUE]	Chargeoff Present [TEMPLATE VALUE]	Behaviour Score [TEMPLATE VALUE]	Aggregation Sub-Score
EXCELLENT	0	None	90 to 100	90
GOOD	1 to 2	None	75 to 89	80
FAIR	3 to 5	Older than 2 years	50 to 74	65
POOR	6 or more	Within past 2 years	25 to 49	30
UNACCEPTABLE	Multiple chargeoffs	Active collections	5 to 24	5

[TEMPLATE VALUE] Behaviour Scoring Bands

EXCELLENT: 0 delinquencies, no chargeoffs — `behaviour_score` = 95 — aggregation sub-score = 90. GOOD: 1 to 2 delinquencies, no chargeoffs — `behaviour_score` = 80 — aggregation sub-score = 80. FAIR: 3 to 5 delinquencies OR chargeoff older than 2 years — `behaviour_score` = 65 — aggregation sub-score = 65. POOR: 6 or more delinquencies OR chargeoff within 2 years — `behaviour_score` = 30 — aggregation sub-score = 30. UNACCEPTABLE: multiple chargeoffs or active collections — `behaviour_score` = 5 — aggregation sub-score = 5. `chargeoff_history` = True if any tradeline has been written off by a lender.

7.9 Income Verification and DTI Policy

ADS node: `income_analysis_node`.

7.9.1 Policy Statement

No loan shall be disbursed without verified income. DTI must be calculated using verified gross monthly income. Applications where DTI exceeds the maximum threshold must be declined or restructured.

DTI Range [TEMPLATE VALUE]	Income Risk	affordability_flag	Policy Action
0% to 30%	LOW	True	Full approval if capacity permits
31% to 40%	MODERATE	True	Approve with income documentation on file
41% to 50%	HIGH	True	Approve; flag for post-disbursement monitoring
Above 50%	UNACCEPTABLE	False	DECLINE or COUNTER_OFFER with reduced principal

[TEMPLATE VALUE] DTI Thresholds and Affordability Rules

DTI = (sum_open_monthly_obligations + proposed_new_EMI) / verified_gross_monthly_income. LOW: DTI 0.00 to 0.30 — income_risk = LOW, affordability_flag = True. MODERATE: DTI 0.31 to 0.40 — income_risk = MODERATE, affordability_flag = True. HIGH: DTI 0.41 to 0.50 — income_risk = HIGH, affordability_flag = True. UNACCEPTABLE: DTI above 0.50 — income_risk = UNACCEPTABLE, affordability_flag = False. Missing income: income_missing_flag = True when monthly_income is None or zero; treat DTI as UNACCEPTABLE. Maximum DTI allowed for any counter-offer option = 0.40.

7.10 RAG Knowledge Base Maintenance Policy

ADS node: rag_retrieval_node (Indian variant workflow).

7.10.1 Policy Statement

The bank maintains a Qdrant vector store containing chunked policy documents. The ADS queries this store at runtime to retrieve applicable thresholds for each risk signal node. This section governs what must be indexed and how retrieval maps to each node.

7.10.2 Documents That Must Be Indexed

Document	Collection Name	Re-index Trigger
This Bank Lending Policy Manual (this document is the source of all other policy documents)	bank_policy	Any threshold change; at minimum annually
RBI Master Direction on Automated Credit Assessment Guidelines	cbi_guidelines	Each new RBI circular
RBI Master Direction on Interest Rate on Advances Guidelines	rbi_guidelines	Each rate direction update
RBI Digital Lending Guidelines	rbi_guidelines	Each guideline revision

7.10.3 Per-Node Query Mapping

ADS Node	Query Topic Sent to Vector Store
credit_score_node	score band thresholds, base limits, risk flags
public_record_node	bankruptcy severity, adjustment factors, hard-decline rules
utilization_node	revolving balance thresholds, utilisation adjustment factors
debt_exposure_node	debt leverage limits, monthly obligation caps
inquiry_node	inquiry velocity thresholds, penalty structures
payment_behavior_node	delinquency severity, chargeoff rules, behaviour score bands
income_analysis_node	DTI thresholds, affordability rules, income verification

- Empty RAG retrieval causes the node to use hardcoded FALLBACK defaults.
- Re-index within 5 business days of any threshold change.
- FALLBACK rate above 5% of daily volume triggers immediate re-index review.

7.11 Risk Aggregation Weights and Tier Policy

ADS node: risk_aggregator_node.

7.11.1 Policy Statement

Seven risk signals are combined into a composite score using a weighted average. Weights are fixed and may only change after a formal Model Governance review approved by the Chief Credit Officer.

Risk Factor	Weight [TEMPLATE VALUE]	ADS Node	Normalisation Rule
Credit Score	0.25	credit_score_node	(score minus 300) divided by 5.5, clamped 0 to 100
Payment Behaviour	0.15	payment_behavior_node	Use raw behaviour_score (0 to 100)
Public Records	0.15	public_record_node	NONE = 100, LOW = 90, MODERATE = 60, SEVERE = 30
Credit Utilisation	0.15	utilization_node	EXCELLENT = 90, GOOD = 60, MODERATE = 65, CRITICAL = 10
Income and DTI	0.15	income_analysis_node	LOW = 90, MODERATE = 60, HIGH = 65, UNACCEPTABLE = 5
Debt Exposure	0.10	debt_exposure_node	LOW = 90, MODERATE = 60, HIGH = 65, EXTREME = 5
Inquiry Velocity	0.05	inquiry_node	LOW = 90, MODERATE = 60, HIGH = 65, CRITICAL = 10

[TEMPLATE VALUE] Aggregation Weights, Formula and Risk Tier Bands

aggregated_risk_score = sum of (normalised_sub_score multiplied by weight) for all 7 factors. Weights sum to 1.00. Hard-decline override: if hard_decline_flag = True then aggregated_risk_score = 0.0. Tier A: aggregated_risk_score >= 80 — APPROVE. Tier B: aggregated_risk_score 65 to 79 — APPROVE with conditions. Tier C: aggregated_risk_score 50 to 64 — COUNTER_OFFER likely. Tier D: aggregated_risk_score 35 to 49 — COUNTER_OFFER or DECLINE. Tier F: aggregated_risk_score below 35 — DECLINE.

7.12 Lending Decision and Pricing Policy

ADS node: `decision_llm_node`.

7.12.1 Decision Protocol

Step	Condition	Action
1	Risk Tier = F OR <code>hard_decline_flag</code> = True	Output DECLINE immediately
2	Assign interest rate by tier	Use rate schedule in 7.12.2
3	Calculate max capacity	$\text{max_approved} = \text{base_limit_band} \times \text{public_record_adj} \times \text{util_adj} \times \text{inquiry_penalty}$
4	<code>affordability_flag</code> = False	Output DECLINE
5	<code>Requested</code> <= <code>max_approved</code>	Output APPROVE at requested amount
6	<code>Requested</code> > <code>max_approved</code> AND <code>max_approved</code> > 0	Output COUNTER_OFFER
7	<code>max_approved</code> = 0	Output DECLINE
8	APPROVE: calculate disbursement	$\text{disbursement_amount} = \text{approved_amount} \text{ multiplied by } 0.98$

7.12.2 Interest Rate Schedule

Tier	Annual Rate [TEMPLATE VALUE]	Monthly Rate	Products [TEMPLATE VALUE]
A (PRIME)	7.5%	0.625%	Personal loans, home loans, MSME term loans
B (NEAR_PRIME)	10.0%	0.833%	Personal loans, vehicle finance
C (FAIR)	13.5%	1.125%	Unsecured personal credit, top-up loans
D (SUBPRIME)	18.0%	1.500%	Secured personal loans, micro-credit only
F (DECLINE)	N/A	N/A	No lending permitted

[TEMPLATE VALUE] Interest Rate and Origination Fee

Tier A interest rate = 7.5% per annum. Tier B interest rate = 10.0% per annum. Tier C interest rate = 13.5% per annum. Tier D interest rate = 18.0% per annum. `origination_fee_rate` = 0.02 (2% of approved amount). `disbursement_amount` = `approved_amount` multiplied by (1 minus `origination_fee_rate`). Example: approved INR 10,00,000 at 2% fee yields disbursement of INR 9,80,000.

7.13 Counter-Offer Generation Policy

ADS node: counter_offer_node.

7.13.1 Policy Statement

When requested amount exceeds assessed capacity, the counter-offer node must generate at least two restructured loan options. Every option must bring the EMI within the affordability ceiling.

Option ID	Principal Reduction [TEMPLATE VALUE]	Tenure Change [TEMPLATE VALUE]	EMI Ceiling	Min Tenure
OPT_REDUCED_AMOUNT	15% reduction (x 0.85)	No change	35% of monthly income	12 months
OPT_EXTENDED_TERM	30% reduction (x 0.70)	Add 12 months	35% of monthly income	36 months

[TEMPLATE VALUE] Counter-Offer Generation Rules

max_affordable_emi = verified_monthly_income multiplied by 0.35. OPT_REDUCED_AMOUNT: proposed_amount = requested_amount multiplied by 0.85, tenure unchanged. OPT_EXTENDED_TERM: proposed_amount = requested_amount multiplied by 0.70, proposed_tenure = requested_tenure plus 12 months. Origination fee 2% applies to disbursement on all options. counter_offer_logic must state reason original amount was not sanctioned. If max_approved = 0 then output DECLINE instead of counter-offer.

7.14 Model Fallback and Human Override Policy

Applicable to all LLM-based ADS nodes.

7.14.1 Fallback Activation

Every LLM-based node must implement a deterministic fallback path that activates when the LLM call fails, times out, or returns an unparseable response. All fallback activations must be logged with `llm_response_type = FALLBACK`.

Node	Fallback Defaults Applied
credit_score_node	Hardcoded score bands and base limits from Section 7.3
public_record_node	Hardcoded severity rules from Section 7.4
utilization_node	Hardcoded utilisation tiers from Section 7.5
debt_exposure_node	Hardcoded exposure tiers from Section 7.6
inquiry_node	Hardcoded velocity tiers from Section 7.7
payment_behavior_node	Hardcoded behaviour rubric from Section 7.8
income_analysis_node	Hardcoded DTI bands from Section 7.9
decision_llm_node	score >= 75 then APPROVE at requested amount and tier rate; score 55 to 74 then COUNTER_OFFER at 80%
counter_offer_node	OPT_REDUCED_AMOUNT at 85% of requested; OPT_EXTENDED_TERM at 70%, plus 12 months

7.14.2 Human Override Requirements

- FALLBACK decisions must be reviewed by a credit officer within 24 hours.
- Human overrides must log: officer ID, timestamp, override reason.
- Override volume above 5% of daily applications triggers Technology Risk escalation.

7.15 Audit, Reporting and Compliance

Applicable to all ADS nodes via audit decorator.

7.15.1 Mandatory Audit Fields

Field	Content	Retention [TEMPLATE VALUE]
application_id	Unique application identifier	Permanent
correlation_id	Request trace identifier	Permanent
node_execution_times	Duration of each node in milliseconds	90 days
model_reasoning	LLM explanation per classification	7 years
llm_response_type	RAG or FALLBACK	7 years
confidence_score	Model confidence 0 to 1 per output	7 years
reasoning_trace	Aggregator sub-scores and hard-decline flag	7 years
reasoning_steps	Decision node step-by-step logic	7 years
rag_pool	Source documents and chunk IDs used	7 years
decision	APPROVE, COUNTER_OFFER, or DECLINE	7 years
decline_reason	Explanation for adverse decision	7 years

7.15.2 Reporting Schedule

- Monthly to Board Risk Committee: approval rate, decline rate, FALLBACK rate, override rate.
- Quarterly model validation to confirm thresholds reflect current portfolio performance.
- Annual third-party model audit if portfolio above INR 500 crore.
- Adverse action notice to applicant within 7 working days of DECLINE or COUNTER_OFFER.

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